

Fermion Mass-Web from Winding Steps — Casey-seeded investigation v0.1

Keeper (capturing Casey interpretive seed 2026-05-28 Thursday)

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Fermion Mass-Web from Winding Steps

Casey's seed (verbatim intent)

On Grace's finding that $m_s/m_d = 2\pi^2$ survives as scheme-invariant (light/light ratio), Casey: > “this seems to say ‘another winding around the circle’ ... we should end up with practical relationships between all fermions and quarks, perhaps all particles.”

The criterion this sharpens (the strong, usable part)

The scheme-invariance audit + Casey's reading converge on a clean criterion:

A fermion ratio is forward content (scheme-invariant) if and only if it is expressible in substrate geometry (π -powers, operational integers, windings). Scheme-*dependent* cross-tier ratios are precisely the ones that LACK clean substrate-geometric form.

This is a much better organizing principle than “scheme-dependent = shelve” (per [[feedback_show_all_threads_then_weave]]). It tells us *where to look* for forward results: the dimensionless, geometric ratios.

Evidence (all scheme-invariant, all substrate-geometric): - $m_s/m_d \approx 2\pi^2$ (1.3%) — $2\pi^2 = \text{vol}(\text{unit } S^3)$, a genuine winding/circumference measure (“another winding”) - $m_\mu/m_e = (24/\pi^2)^6$ (0.00%, T190) — geometric, high π -power - $m_\tau/m_e = 49.71 = g^2 \cdot \text{Ogg71}$ (0.05%, T2003) — operational-integer - mixing angles (Cabibbo 9/40, PMNS 42/137 etc.) — dimensionless, scheme-invariant by construction - $m_W/m_Z = \sqrt{g/N_c}$ (0.05%) — geometric

What does NOT hold uniformly (honest brake — keep the thread live)

The *specific* “small integer $\times \pi$ -power” winding form (Casey's literal “winding around the circle”) fits m_s/m_d but NOT uniformly: - m_s/m_d : best $n \cdot \pi^2 = 2 \cdot \pi^2$ (1.3%) — $m_\tau/m_\mu \approx 16.82$: best $n \cdot \pi^2 = 5 \cdot \pi$ (6.6%) $\times$ - $m_\mu/m_e \approx 206.8$: best $n \cdot \pi^2 = 7 \cdot \pi^3$ (5.0%) $\times$ — but $(24/\pi^2)^6$ nails it (0.00%)

So the ratios are geometric in several different ways (single windings like $2\pi^2$, high π -powers like $(24/\pi^2)^6$, operational integers like 49.71). The web exists; it is not a single uniform winding form. Keep both threads live: (a) simple-winding edges, (b) higher π -power / operational-integer edges.

The program Casey's vision points to (the real next step)

If every scheme-invariant ratio is a product of winding steps, then a small spanning set of windings determines ALL fermion masses up to one overall scale — “practical relationships between all fermions and quarks.” This is exactly the Winding-Composite Generations (WCGP) territory: Grace's multiplicative chains (<0.13% across 3 sectors) ARE candidate web-edges.

Testable structural claim: the scheme-invariant fermion ratios form a connected, compositional web with few generators. Test by composition closure, e.g.: - $m_\tau/m_e = ?$ (m_τ/m_μ) $\cdot$ (m_μ/m_e) — does the product of

geometric edges reproduce the geometric whole? - Span: how few “winding steps” generate all 9 charged-fermion masses + 3 neutrino splittings?

Investigation routing

- Grace: catalog EVERY scheme-invariant fermion ratio (lepton, up-type same-scale, down-type same-scale, neutrino splittings); express each in substrate geometry; test web-connectivity via composition (do edges compose to the whole?). Build the web graph.
- Lyra: mechanism — why winding steps? which geometric edge-type does each sector use (single-winding vs π -power vs operational-integer), and is that region-correlated (Shilov vs bulk)? Connect to WCGP + the winding-mode axis W_0 .
- Elie: numerics — for each ratio, which geometric form fits best + at what precision; test web closure (spanning-set $\rightarrow$ all masses); quantify how many generators.

Honest tier disposition

- Criterion (scheme-invariant ∇ substrate-geometric): STRONG, testable, organizing principle. FRAMEWORK-PLUS.
- $m_s/m_d = 2\pi^2$ as a winding: clean single data point; geometric reading sound; 1.3% (not $<1\%$).
- Complete fermion mass-web from few winding generators: PROGRAM / hypothesis, not a result. Pursue per Casey; weave the story after the web is mapped.
- Do NOT claim “all fermion masses derived” until the web closes with a spanning set AND the edges are scheme-invariant AND a mechanism explains the winding steps. Cal #27 applies; but per Casey, keep every thread live and investigate fully.

— Keeper, 2026-05-28 Thursday. Casey-seed captured; routed to Grace/Lyra/Elie; criterion strong, web is the program.

Addendum — Casey’s invariant-anchor refinement (the MECHANISM behind the criterion)

Casey: “windings occur around other geometric objects ... the area of a circle [is] the volume of S^3 which is interesting in itself. Geometric or topological ‘anchors’ are invariants.”

This supplies the mechanism the criterion was missing. It is no longer just “scheme-invariant ∇ geometric” as an empirical pattern — there is a *reason*:

A physical ratio is scheme-independent if and only if it is anchored to a topological/geometric INVARIANT of a substrate geometric object. Invariants are frame-independent by their nature; a ratio equal to one inherits that independence. A scheme-dependent ratio is, by contradistinction, NOT anchored to an invariant — it is a frame artifact.

This is clean, it is the right “why,” and it makes the criterion a theorem-shaped statement rather than an observation.

Precise identities (accuracy note — Keeper)

$2\pi^2$ is the volume of the unit S^3 , not the area of a disk (disk = π). But Casey’s reading decodes exactly: $2\pi^2 = (2\pi) \cdot \pi = \text{circumference}(S^1) \times \text{area}(\text{disk})$ — “a winding around the circle times the enclosed disk.” His “winding around the circle” framing is literally correct once the second factor is named. Substrate menu: - circumference $S^1 = 2\pi \cdot |S^2 = 4\pi \cdot |S^3 = 2\pi^2 \cdot |S^4 = 8\pi^2/3 \cdot |S^5 = \pi^3 \cdot | \text{disk } B^2 = \pi$

Substrate anchoring (why these objects, not arbitrary spheres)

D_{IV}^5 ’s Shilov boundary is $S^4 \times S^1$, and $S^3 \subset S^4$. So $2\pi^2 = \text{vol}(S^3)$ is an intrinsic invariant of the substrate’s own boundary geometry — $m_s/m_d \approx 2\pi^2$ is a winding/measure on the substrate boundary, not a coincidence with

an arbitrary sphere. Each candidate web-edge must anchor to a specific invariant of a specific substrate object.

The two anchor classes (both scheme-free)

1. Metric invariants (carry π): sphere volumes ($S^1 \dots S^5$), ball volumes, Bergman/heat-kernel π -powers. These give the π -bearing edges ($2\pi^2$, $(24/\pi^2)^6$, ...).
2. Topological invariants (pure integers, the cleanest): Coxeter $h(B_2)=4$, dual Coxeter $h^\vee(B_2)=3=N_c$, B_2 exponents $\{1,3\}$ / degrees $\{2,4\}$, Q^5 Chern numbers (all BST-primary, $c_5=C_2=6$), Euler characteristics, homotopy/winding numbers. These give the integer edges (49·71, mixing-angle rationals, the Coxeter generation/color counts).

Sharpened routing instruction for the team

For EACH scheme-invariant ratio, don't just write a geometric formula — name the specific invariant and the specific substrate object it anchors to (e.g. “ $m_s/m_d = \text{vol}(S^3 \subset \text{Shilov } S^4)$ ”). An edge without a named invariant anchor is still a lead, not forward content. The web's edges are *labeled by invariants*; that labeling is the forward-derivation, because it explains the scheme-independence rather than merely exhibiting it.

This is a Cal #28 META instance (Casey-interpretive-prompt cascade): the seed “windings around geometric objects” + “anchors are invariants” converted the criterion from pattern to mechanism.

— Keeper, 2026-05-28 Thursday (addendum). Criterion now mechanism-grounded: invariant-anchored  scheme-free.